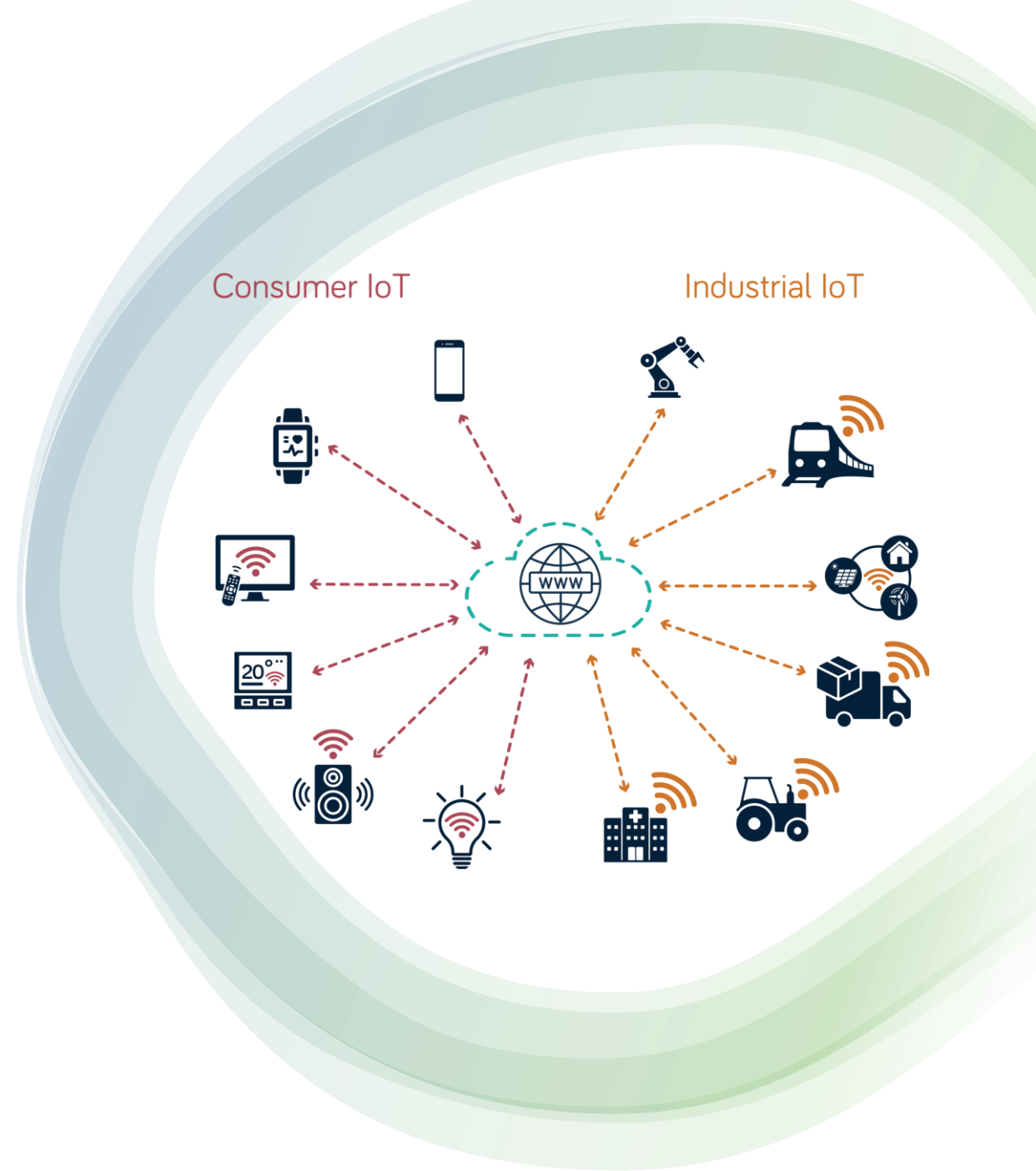


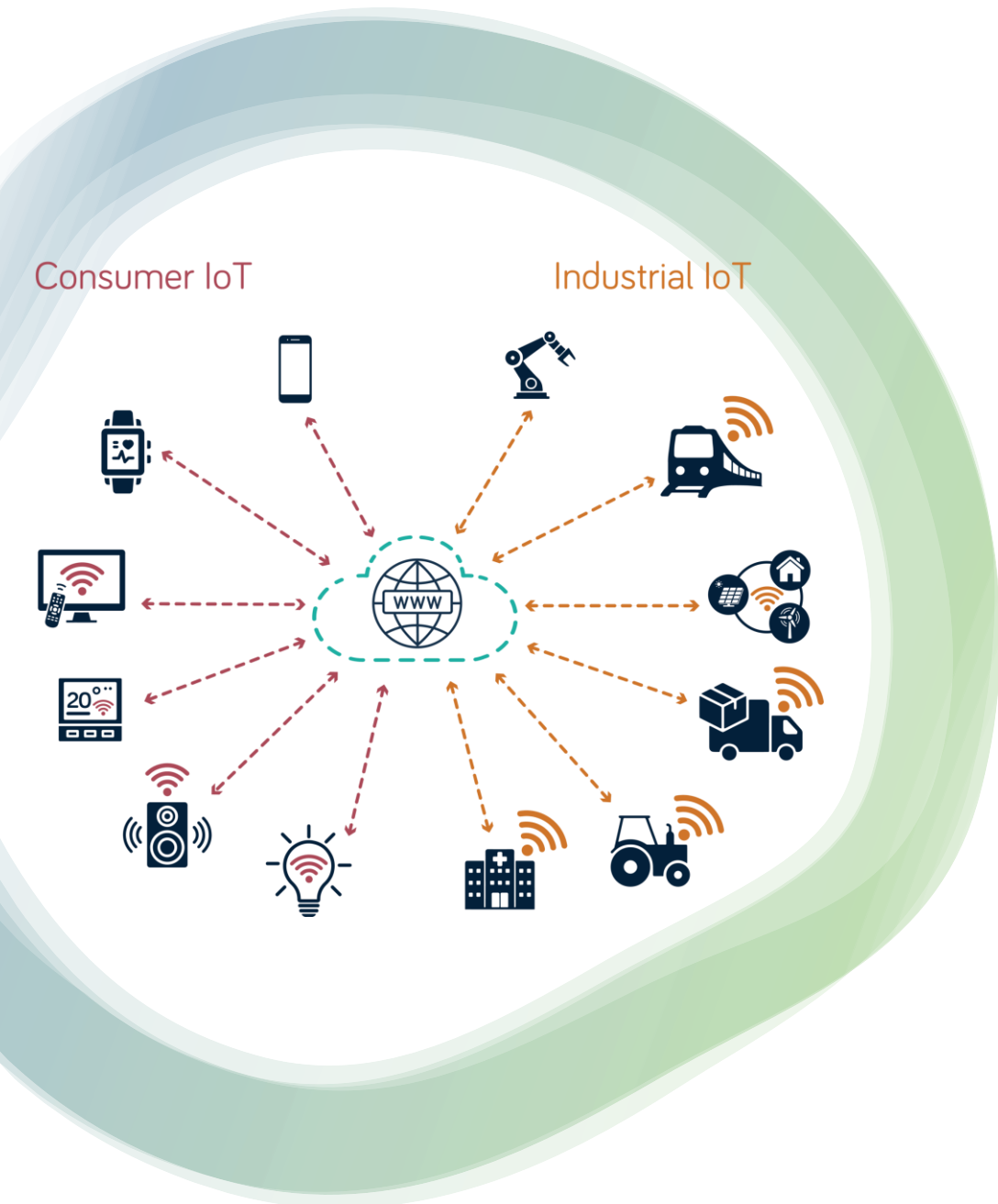
Practical IoT Architectures

Best Practices – Considerations - Advices

(Consumer) IoT

- Network of connected physical devices
- Devices collect, process, and exchange data
- **Focus on convenience and user experience**
- Often consumer-oriented
- **Short to medium product lifecycle**
- Cost-driven (low device and connectivity cost)



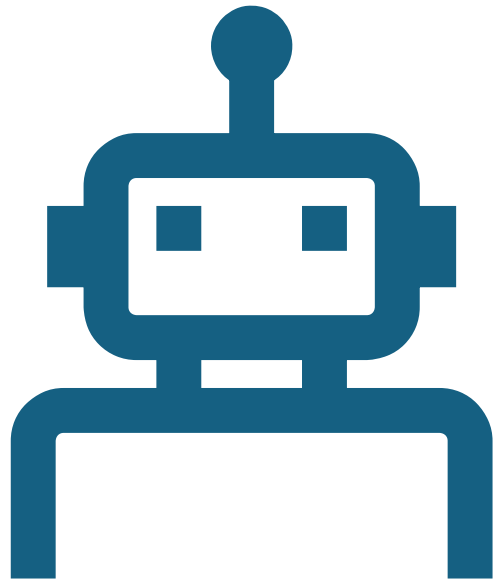


Industrial IoT

- IoT applied in industrial and infrastructure environments
- Devices are part of critical systems and processes
- Focus on **reliability and operational continuity**
- Business- and infrastructure-critical
- Long product lifecycle (**10–20+ years**)
- Integration into existing (legacy) systems

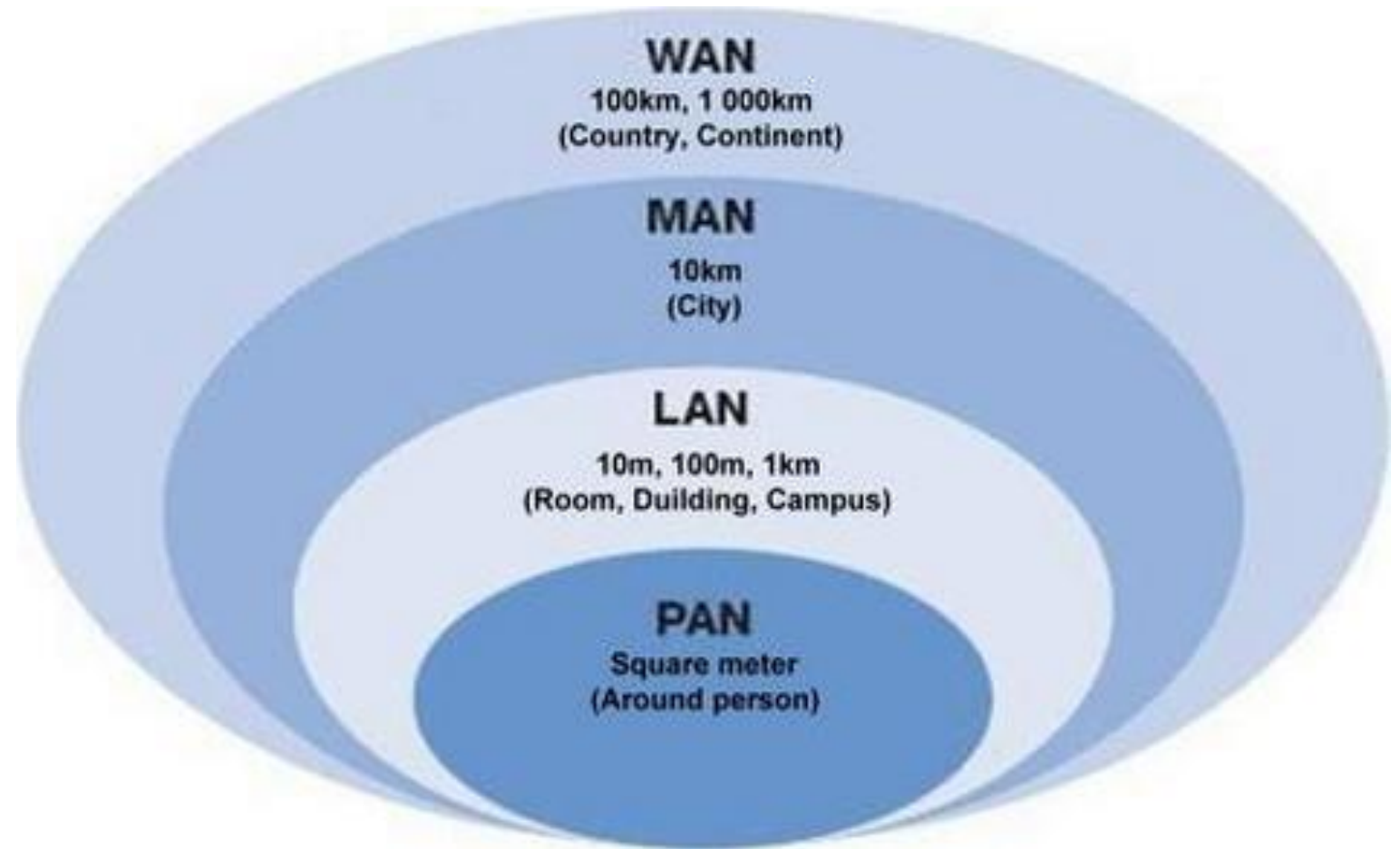
IoT vs IIoT

Aspect	IoT	IIoT
Focus	User convenience	Operational reliability
Failure impact	Low	High
Lifecycle	Short	Long
Cost driver	Hardware cost	Downtime cost



What Types of
Connectivity Do
You Already
Know?

Categories of IoT Connectivity



Bluetooth Low Energy (BLE)



Bluetooth Low Energy (BLE)

Personal Area Network (PAN)

Typical Use Cases:

- Wearables and sensors
- Device configuration
- Smart home applications

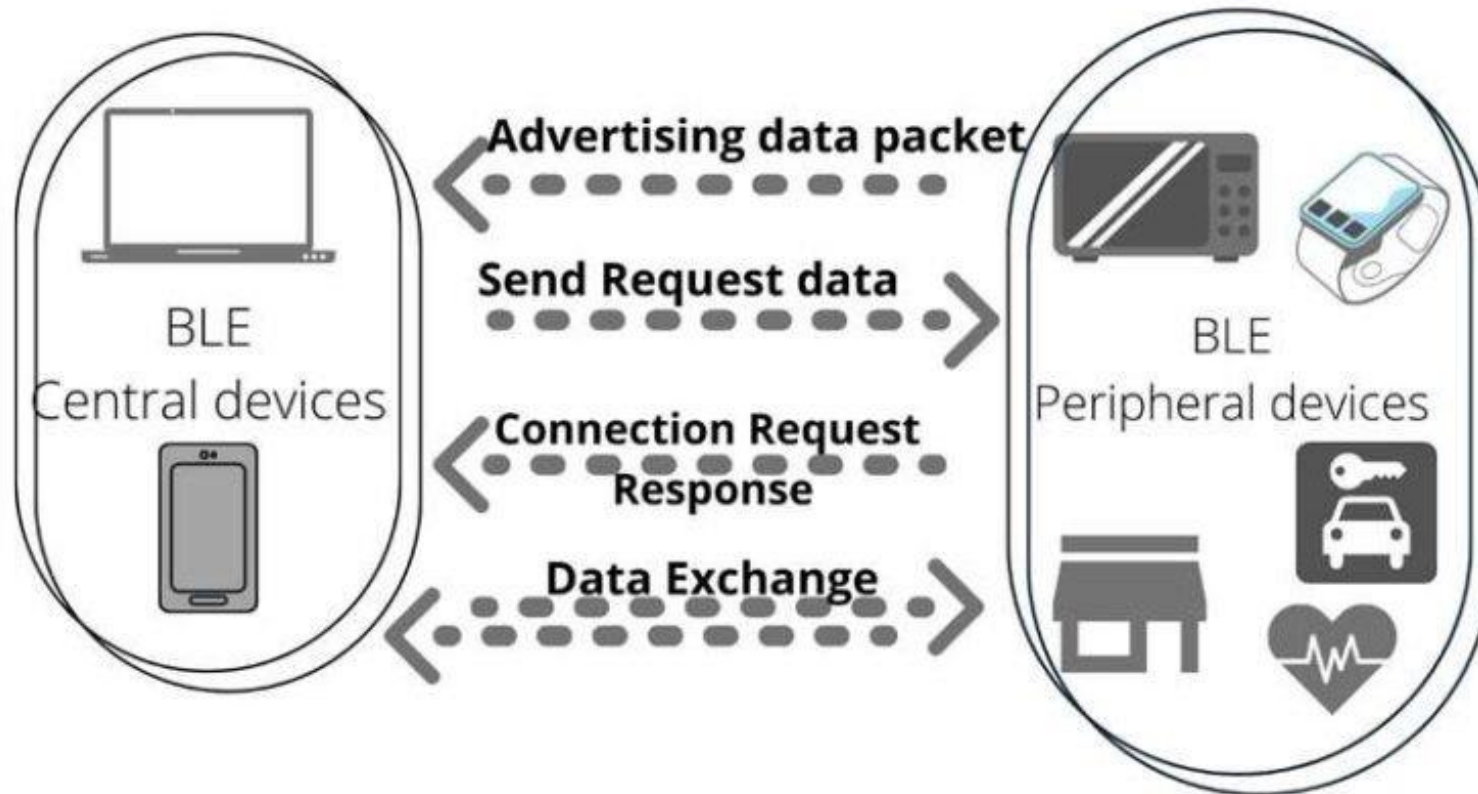
Limitations:

- Short range
- No direct internet connectivity
- **Depends on gateway device** (e.g., smartphone)
- **No guaranteed permanent connection**

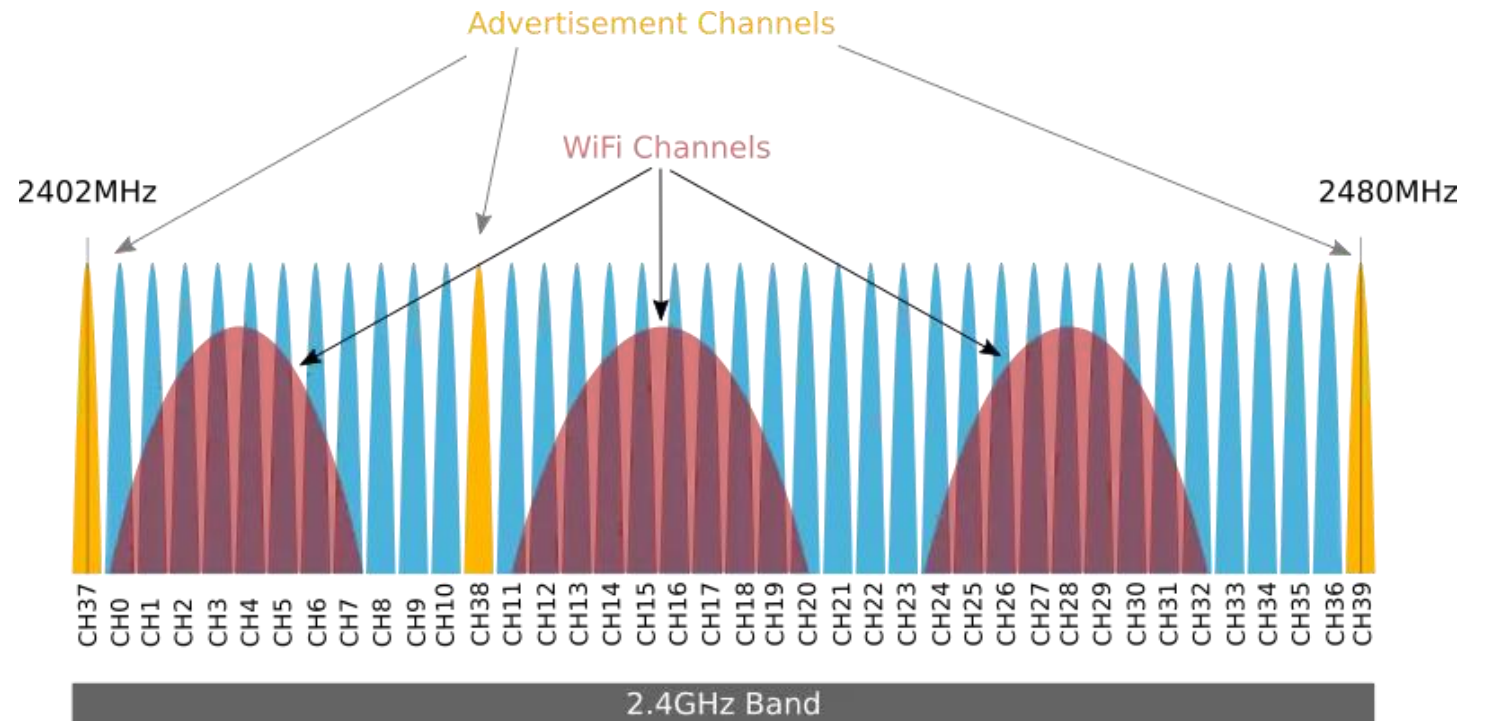
Note: BLE ≠ classic Bluetooth (headphones etc.)

Bluetooth Low Energy (BLE)

- Operates in the 2.4 GHz ISM spectrum,



Bluetooth Low Energy (BLE)



Bluetooth Low Energy (BLE)

128-bit UUID:

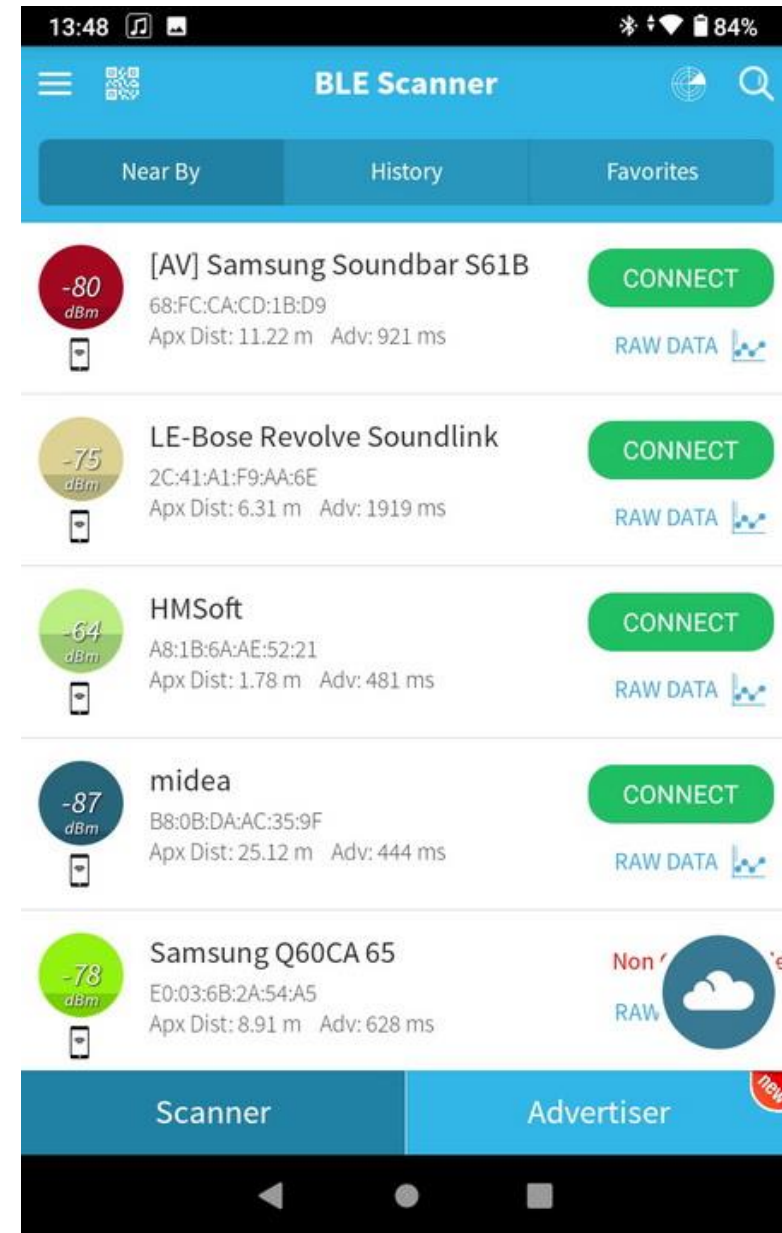
0000180D-0000-1000-8000-00805F9B34FB

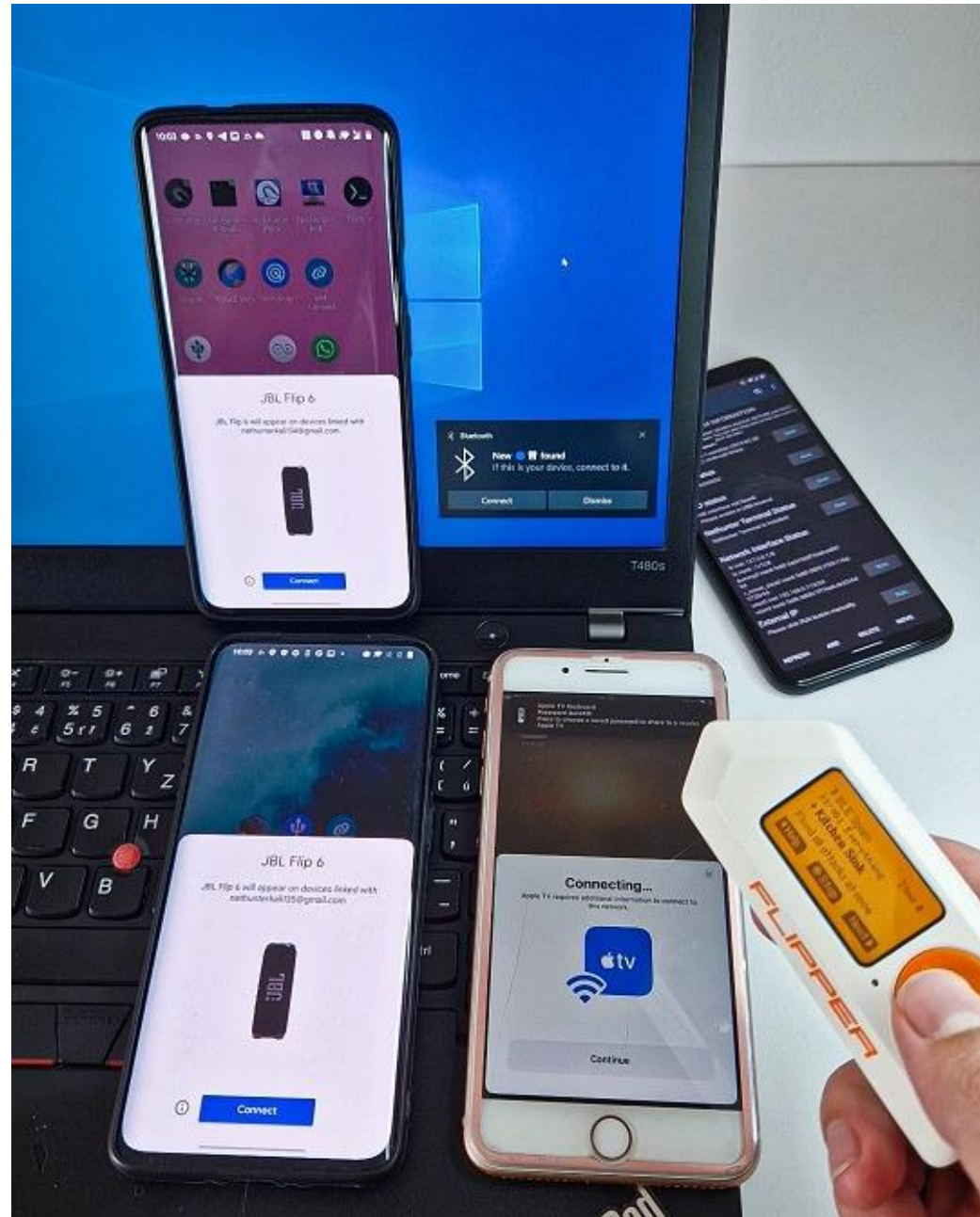
0x180D → Heart Rate Service

0x180F → Battery Service

0x2A19 → Battery Level Characteristic

BLE Advertisement





**Wi-Fi /
Ethernet**



Wi-Fi / Ethernet

Local network (LAN)

Typical Use Cases

- Buildings and homes
- Industrial halls / factories
- **Stationary devices**

Limitations:

- Requires local network access
- **Dependent on customer infrastructure**
- Security and IT restrictions (especially in industry)
- Ethernet requires cabling (but is more stable)



Consumer Cellular (LTE / 5G)

Wide-area network (WAN)

Typical Use Cases

- Consumer devices (smartphones, tablets)
- Devices using customer SIM cards
- Mobile and portable IoT devices

Limitations:

- No centralized control
- **Dependency on customer contracts**
- Not suitable for industrial use cases

Commercial Cellular LTE / 5G



Commercial Cellular

Wide-area network (WAN)

Concept:

- Managed cellular connectivity for IoT devices
- Dedicated SIM cards for products - not users

Central control over:

- SIMs
- Data usage
- Connectivity lifecycle

Commercial Cellular LTE / 5G



Commercial Cellular LTE / 5G

Wide-area network (WAN)

Typical Use Cases:

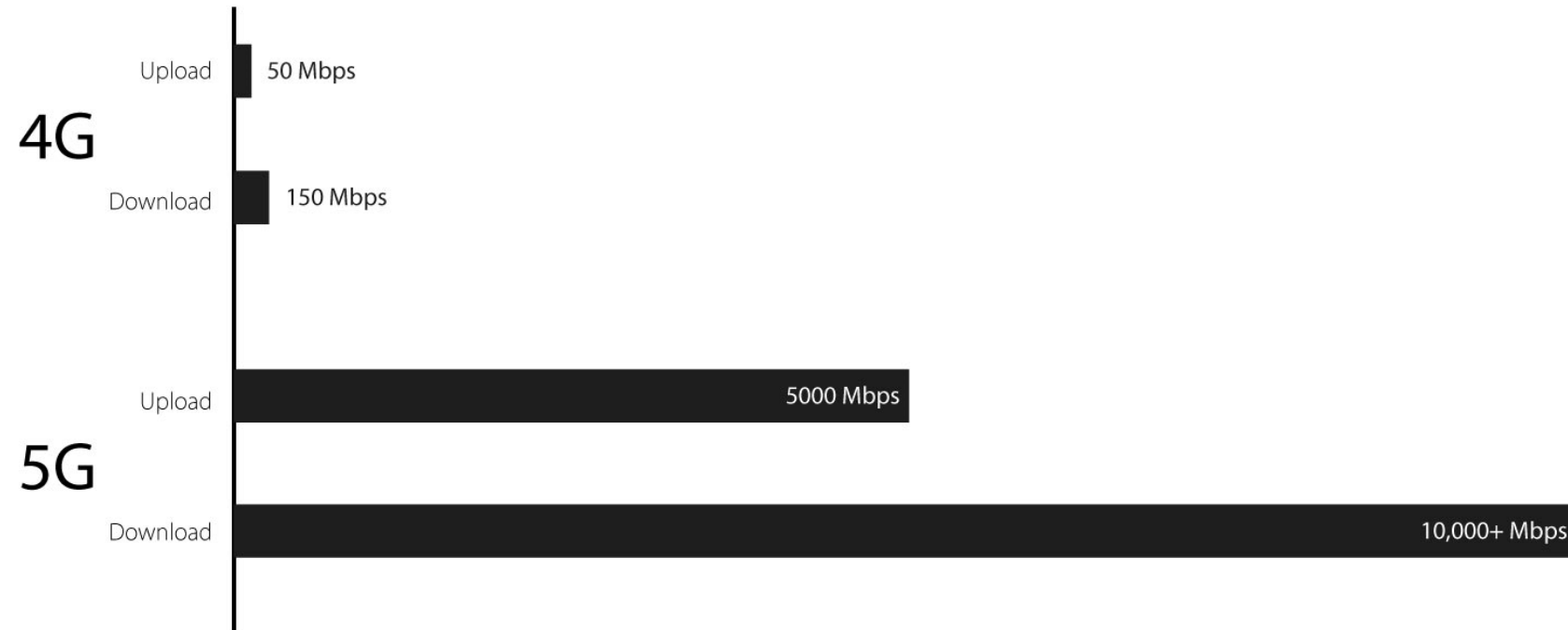
- High data rate applications
- Gateways and routers
- Video transmission
- Remote maintenance

Limitations:

- Higher power consumption
- Higher module and data costs
- Overkill for small sensor data

4G vs 5G Speed

Speeds provided are theoretical maximum speeds.



Standard Lat/Long GPS Coordinate:

48.8376924,10.0734932 → 21 Bytes

Private 5G in Industrial Environments



Private 5G in Industrial Environments

Wide-area network (WAN)

Local private network

Concept:

- Dedicated 5G network in a factory
- Operated by company or provider

Limitations:

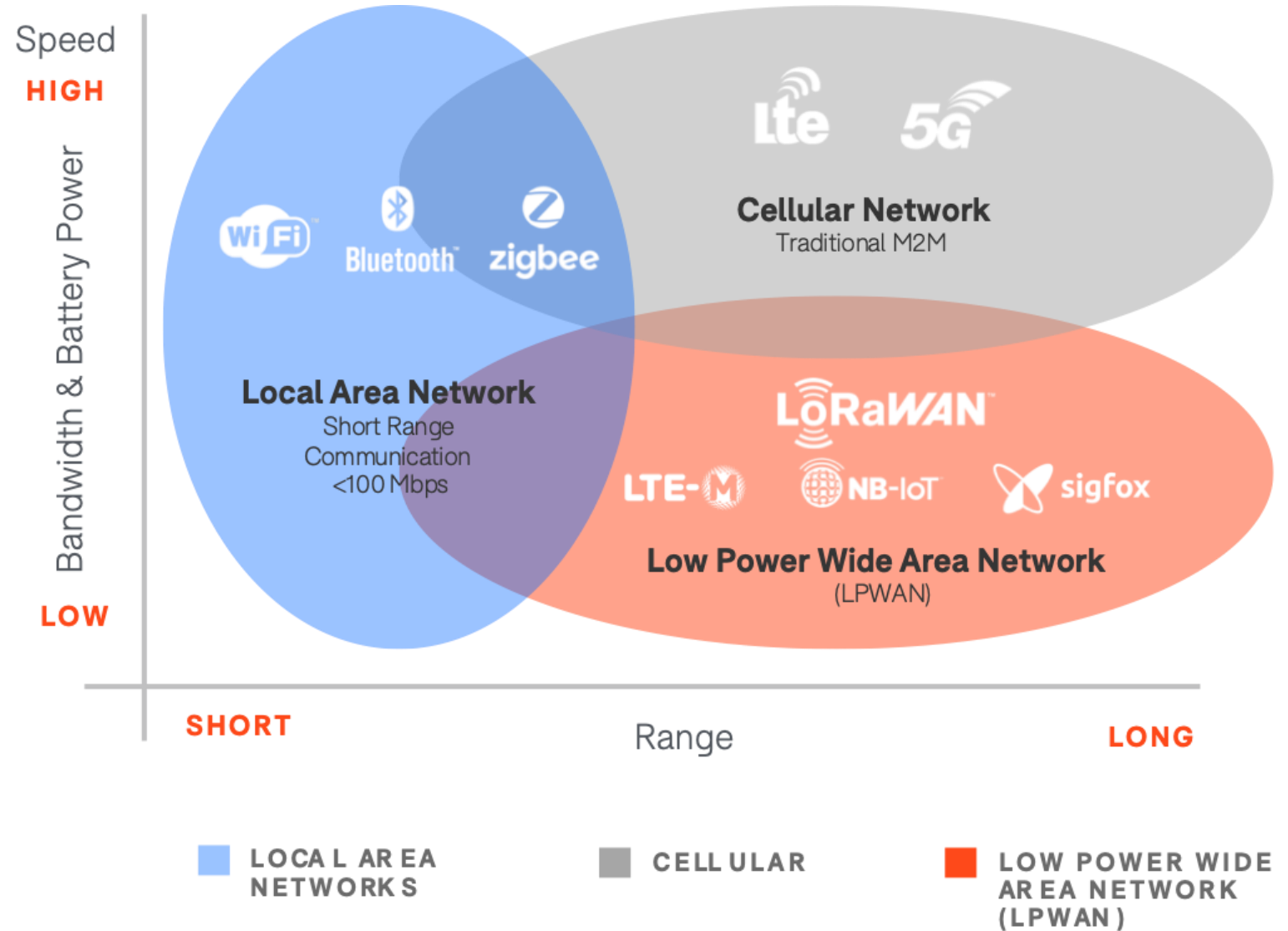
- High deployment cost
- Complex setup and operation
- Limited to specific locations

Private 5G in Industrial Environments



Inline quality checks using high-resolution images require **high throughput and low latency** for the robots.

Low Power Wide Area Networks (LPWAN)



LTE-M

Category:

- Wide-area network (WAN) connectivity
- Low-power cellular

Typical Use Cases:

- Mobile IoT devices
- Asset tracking
- Wearables (cellular)
- Devices with periodic data transmission

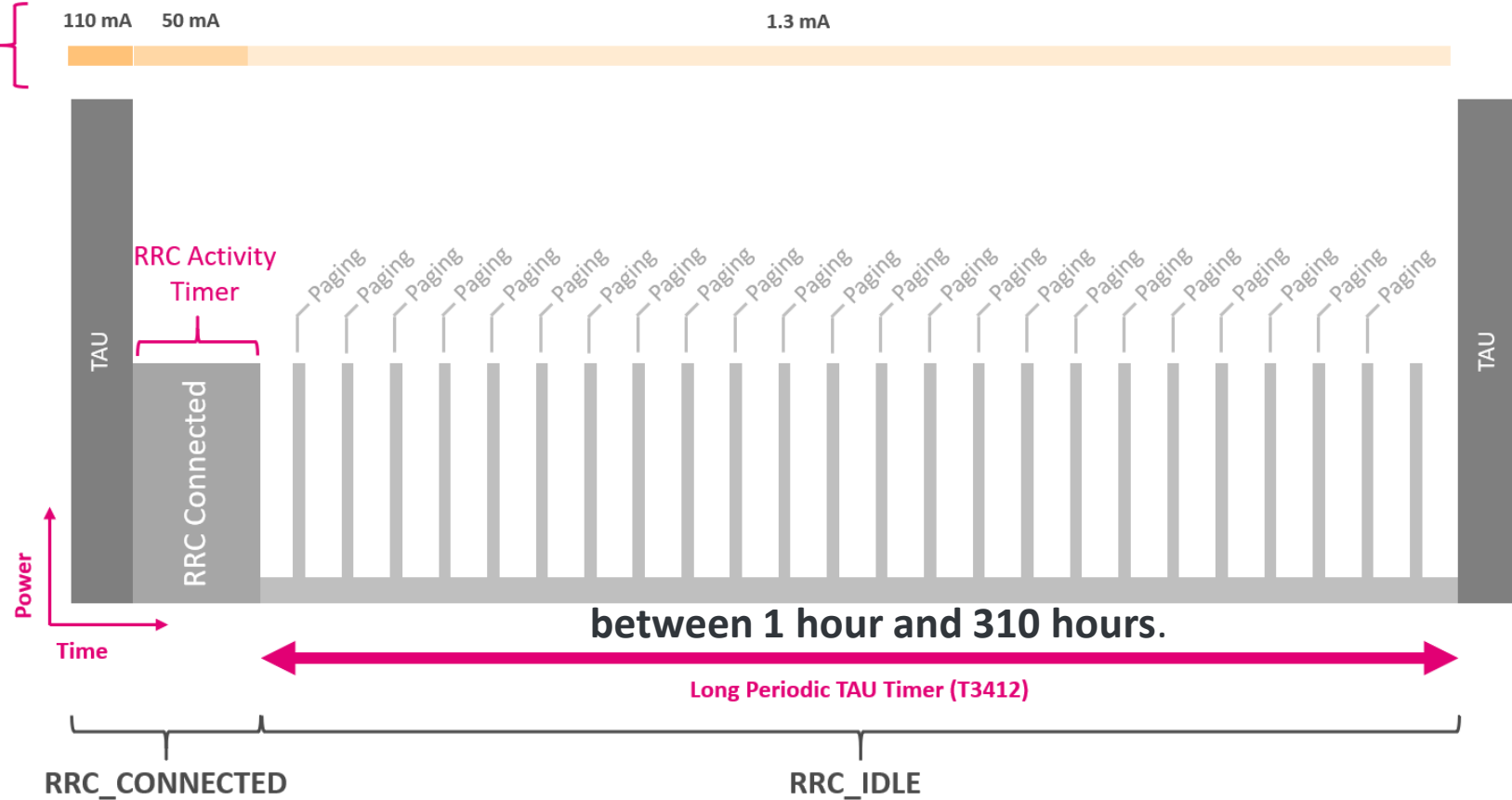
Limitations:

- Still higher power consumption than NB-IoT
- Higher cost compared to LPWAN alternatives
- Not optimized for very small data payloads

LTE-M Power-Saving Features

Long-Periodic Tracking Area Updates (Long-Periodic TAU)

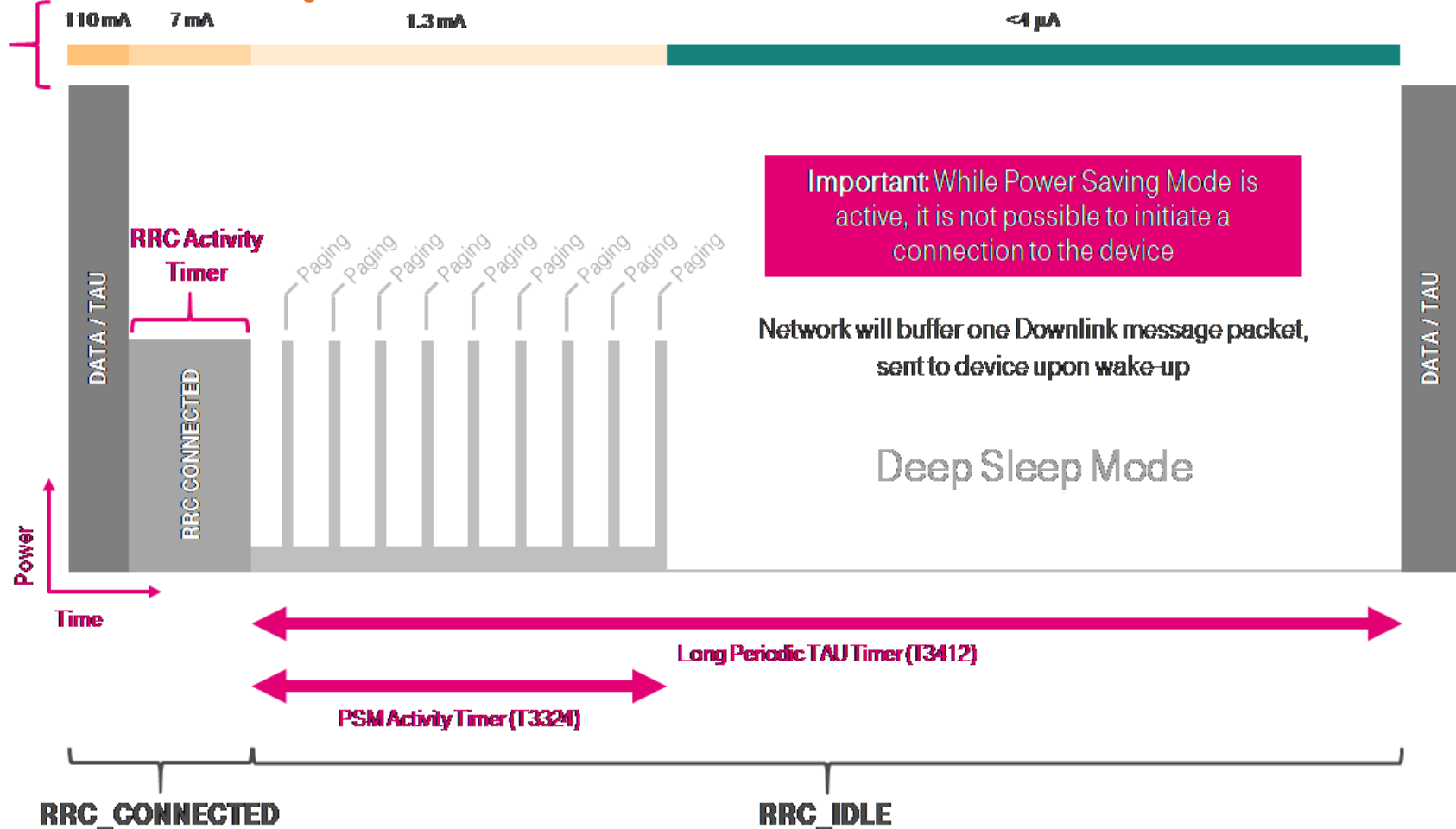
Indicative avg.
power consumption
shown for CE-Level 0



LTE-M Power-Saving Features

Long-Periodic Tracking Area Updates (Long-Periodic TAU)

Indicative avg. power consumption shown for CE-Level 0



NB-IOT



NB-IoT (Narrowband IoT)

Category:

- Wide-area network (WAN) connectivity
- Low-power cellular

Typical Use Cases

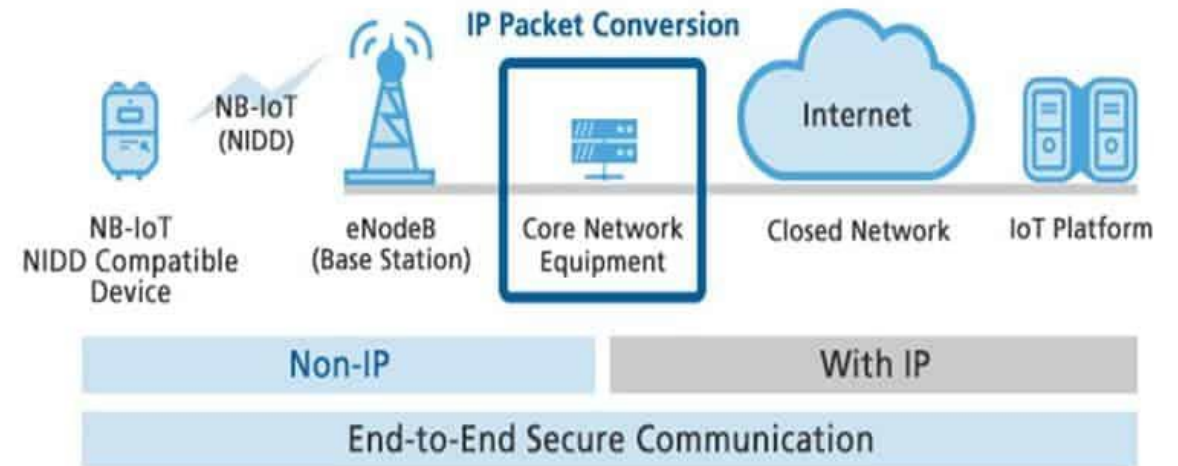
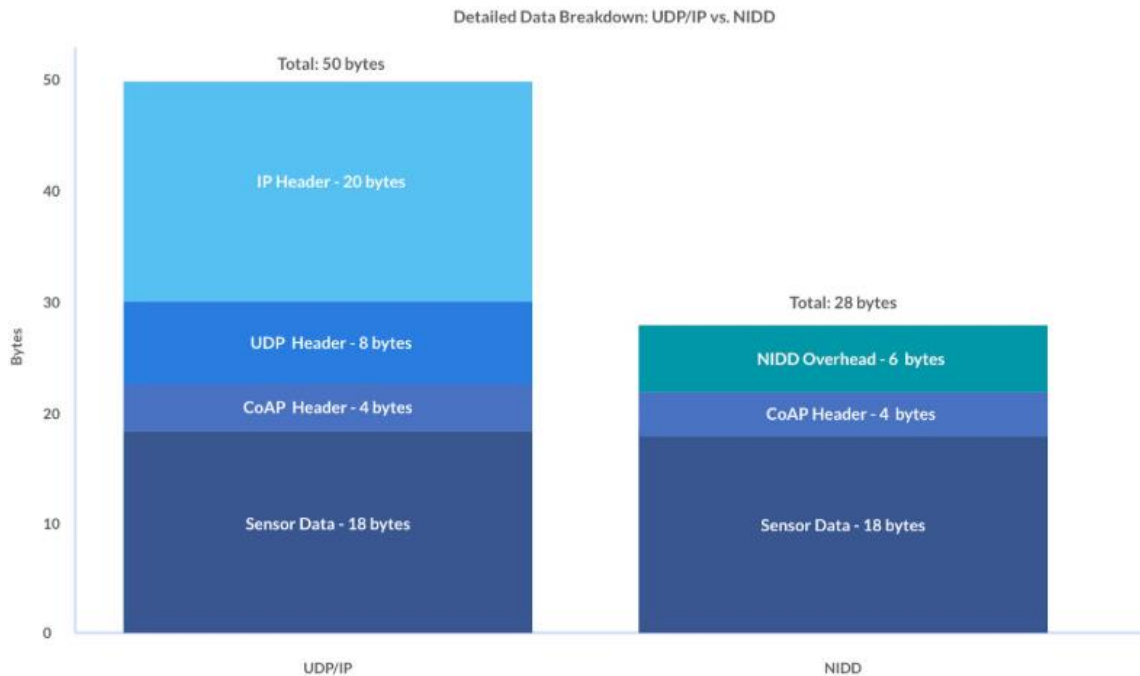
- Small data transmission
- Smart metering (e.g., energy meters)
- Environmental sensors
- Devices in hard-to-reach locations

Limitations

- Very low data rates
- Limited mobility support
- Higher latency possible
- Coverage depends on network provider

NON IP Data delivery (NIDD)

Power consumption can be optimized using non-IP data delivery, as the payload size and transmitted data volume are reduced.



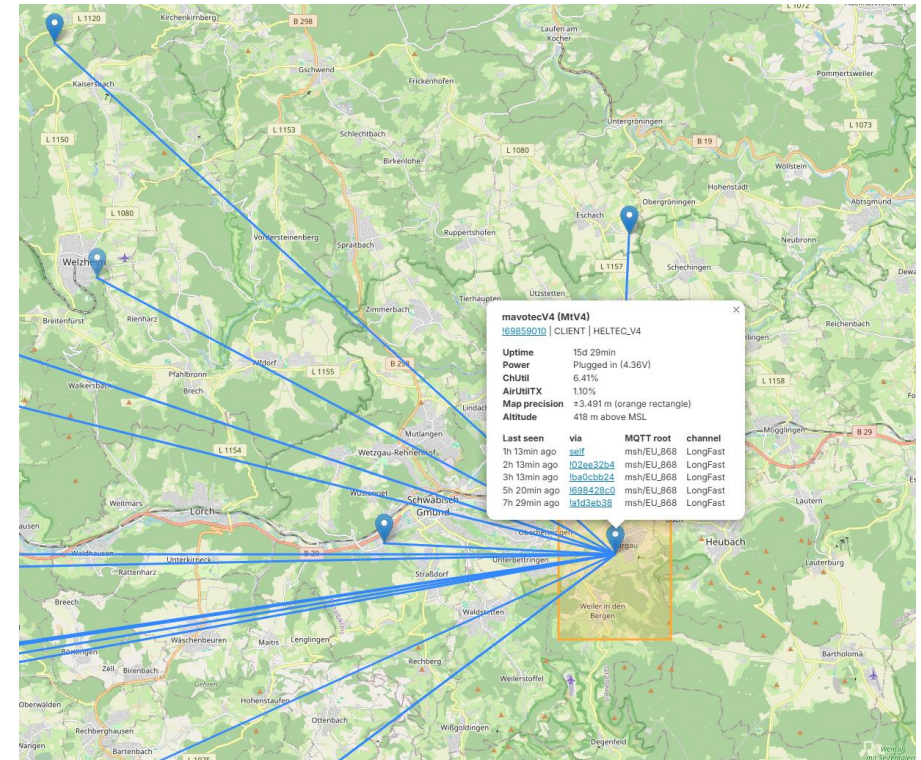
Example: E-scooters using NB-IoT & LTE

- E-scooters can benefit from multiple connectivity technologies: NB-IoT is used for periodic location and battery status updates, while LTE is used during rides for real-time data streaming.



LoRa

<https://meshmap.net/#1770360848>



LoRa / LoRaWAN

Category:

- Wide-area network (WAN) connectivity
- Non-cellular LPWAN

Typical Use Cases:

- Very small data transmission
- Long battery lifetime devices
- Smart cities (municipal infrastructure)
- Agriculture and environmental monitoring
- Campus and building networks

450 MHz



450 MHz Networks

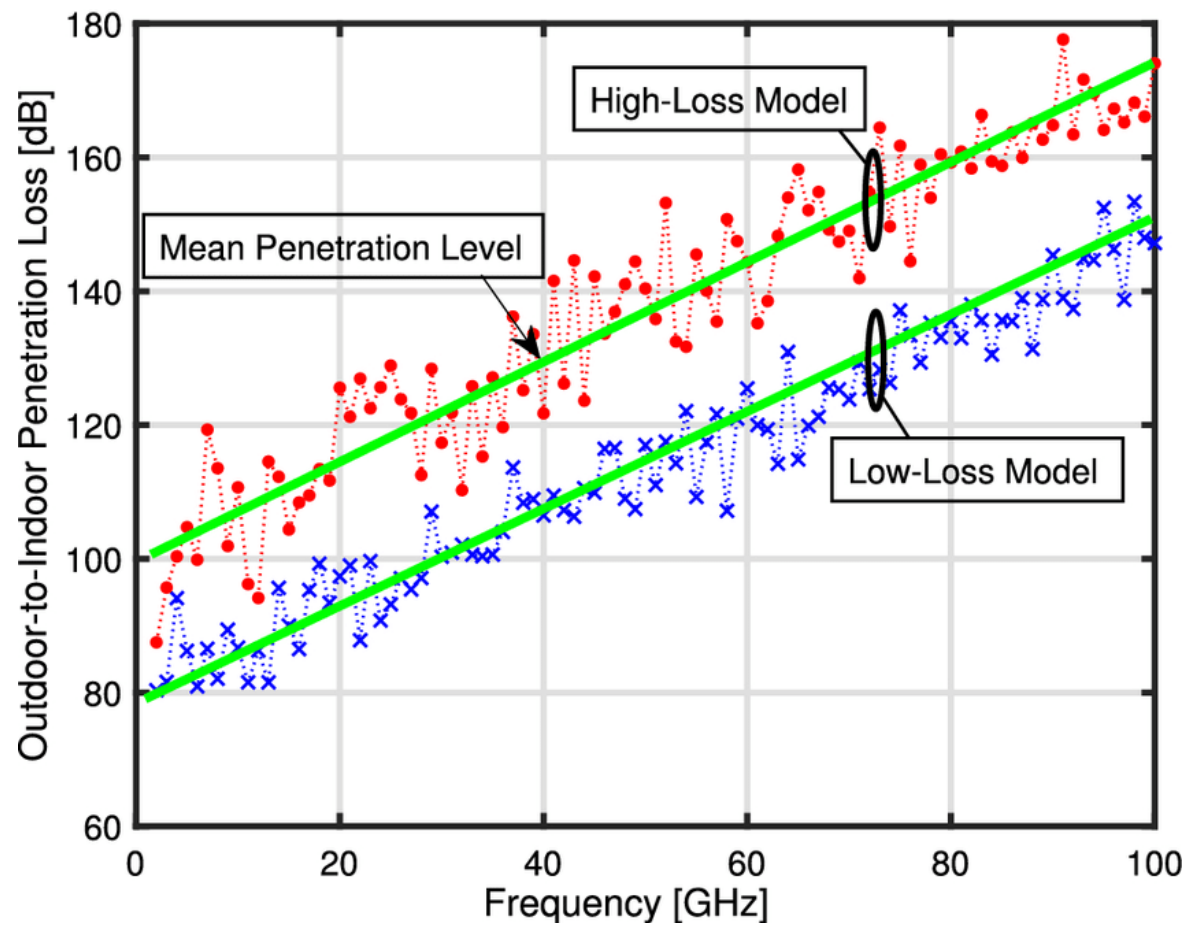
Category:

- Wide-area network (WAN) connectivity
- Specialized cellular / critical infrastructure

Typical Use Cases:

- Energy and utility infrastructure
- Smart grids and metering systems
- Critical infrastructure communication

Why LPWAN?



V·T·E [13][14]	LTE Cat 1	LTE Cat 1 bis	LTE-M				NB-IoT		EC-GSM-IoT
			LC-LTE/MTCe	eMTC					
			LTE Cat 0	LTE Cat M1	LTE Cat M2	non-BL	LTE Cat NB1	LTE Cat NB2	
3GPP release	Release 8	Release 13	Release 12	Release 13	Release 14		Release 13	Release 14	Release 13
Downlink peak rate	10 Mbit/s		1 Mbit/s		~4 Mbit/s		26 kbit/s	127 kbit/s	474 kbit/s (EDGE) 2 Mbit/s (EGPRS2B)
Uplink peak rate	5 Mbit/s				~7 Mbit/s		66 kbit/s (multi-tone) 16.9 kbit/s (single-tone)	159 kbit/s	
Latency	50–100 ms		not deployed	10–15ms			1.6–10 s		700 ms – 2 s
Number of antennas	2	1							1–2
Duplex mode	Full duplex		Full or half duplex				Half duplex		
Device receive bandwidth	1.4–20 MHz			1.4 MHz	5 MHz		180 kHz		200 kHz
Receiver chains	2 (MIMO)	1 (SISO)							1–2
Device transmit power	23 dBm			20 / 23 dBm				14 / 20 / 23 dBm	23 / 33 dBm

Technology	Category	Data Rate	Power Consumption	Range	Mobility	Infrastructure	Typical Use Case
BLE	Short-range	Very low	Very low	Very short	No	Smartphone / Gateway	Wearables, sensors
Wi-Fi / Ethernet	LAN	High	Medium–High	Building	Limited	Local network required	Smart home, machines
Consumer LTE	WAN	High	High	Wide	Yes	Customer SIM	Smartphones, simple IoT
LTE / 5G (Commercial)	WAN	Very high	High	Wide	Yes	Managed SIM	Video, gateways, routers
LTE-M	WAN / LPWAN	Medium	Medium	Wide	Yes	Operator network	Tracking, mobile IoT
NB-IoT	WAN / LPWAN	Low	Very low	Wide	Limited	Operator network	Smart meters, sensors
LoRa	WAN / LPWAN	Very low	Very low	Wide	Limited	Own gateways	Smart city, agriculture
450 MHz	WAN (specialized)	Low	Low	Very wide	Limited	Dedicated network	Energy, infrastructure



How Would You Connect a Smart- Meter?

BPL (Broadband over Powerline)

Hybrid-Category:

- Wired connectivity (using existing infrastructure)
- External Gateway with Cellular Connectivity

Typical Use Cases:

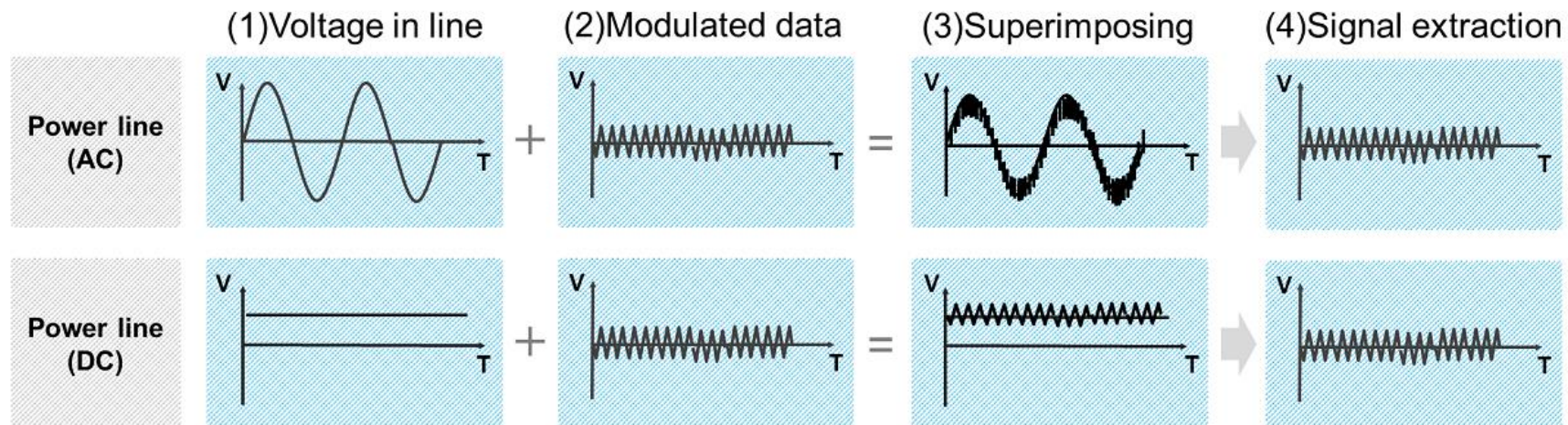
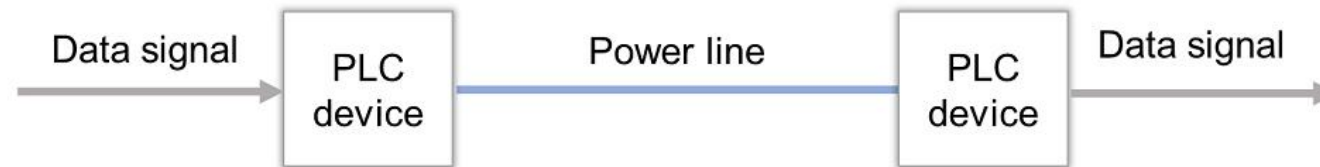
- Smart metering
- Energy infrastructure
- Buildings with difficult radio conditions

BPL for Smart Metering



How Does BPL Work?

Data signal is modulated onto the electrical signal



Stakeholder

IoT Device with SIM



Manufacturer / Solution Provider



Objectives:

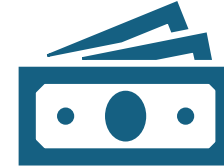
Access to device data (business models)

Remote monitoring and maintenance

Product improvement

Recurring revenue streams

Customer retention



Challenges / Trade-offs:

Desire for maximum data access

Connectivity costs

Data privacy (e.g., GDPR)

Operational complexity

Customer / End User

Objectives:

- Easy and intuitive usage
- Reliable product functionality
- Minimal setup and configuration
- Data privacy and trust
- Low / No ongoing costs

Challenges / Trade-offs:

- Wants convenience and simplicity
- But may reject constant data transmission
- Additional connectivity costs

Cellular Module Manufacturer

Objectives:

- Sell certified communication modules
- Ensure compliance and certifications
- Provide long-term availability
- Offer firmware support and updates

Challenges / Trade-offs:

- Not developed in-house → dependency on supplier
- Long-term availability vs. technology evolution
- Hardware decisions impact entire product lifecycle

Cellular Module Certification

- RF
 - EN 301 908 (LTE/5G)
 - EN 301 511 (GSM)
- EMV
 - EN 301 489-1
 - EN 301 489-52 (Cellular)
- Security
 - EN IEC 62368-1

L 153/62

EN

Official Journal of the European Union

22.5.2014

**DIRECTIVE 2014/53/EU OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL
of 16 April 2014**

**on the harmonisation of the laws of the Member States relating to the making available on the
market of radio equipment and repealing Directive 1999/5/EC**

(Text with EEA relevance)

THE EUROPEAN PARLIAMENT AND THE COUNCIL OF THE EUROPEAN UNION,

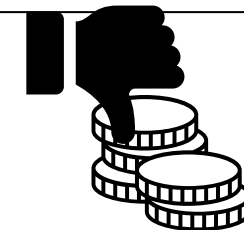
Having regard to the Treaty on the Functioning of the European Union, and in particular Article 114 thereof,

Having regard to the proposal from the European Commission,

After transmission of the draft legislative act to the national parliaments,

Having regard to the opinion of the Economic and Social Committee ⁽¹⁾,

Acting in accordance with the ordinary legislative procedure ⁽²⁾,



Too expensive.
Usually, a provider
solution is preferred.

Certified Modules



[IoT Hardware Ecosystem](#)[Hardware](#)[Services](#)[Wissen](#)[Partner](#)[Kontakt](#)

 [Anmeldung](#)

Module - 3G, 4G
FM101-GL-12



Fibocom

— HIGHLIGHTS —

3G/LTE Cat.12 M.2 module based on the Qualcomm SDX12-0 chipset
Supports a wide range of supported 3G/LTE...

Datenblatt

Module - 3G, 4G
FM101-GL-00



Fibocom

— HIGHLIGHTS —

3G/LTE Cat.6 M.2 module based on the Qualcomm SDX12-2 chipset
Supports a wide range of supported 3G/LTE...

Datenblatt

Module - NB-IoT, LTE-M
MA510-GL-01



Fibocom

— HIGHLIGHTS —

Dual-mode for global market: LTE Cat-M1/Cat-NB2
Reliable & stable data transmission with ultra-lo...

Datenblatt

Module - 2G, 4G
MC610-EU



Fibocom

— HIGHLIGHTS —

Support multiple network frequency bands and rich Internet protocols
Rich extension of interfaces allowing much...

Datenblatt



[Startseite](#) | [Telekom IoT Hardware](#)

Mobile Network Operator

Objectives:

- Provide reliable network connectivity
- Sell connectivity services (SIMs, data plans)
- Expand and utilize network infrastructure

Challenges / Trade-offs:

- Coverage varies by location
- Dependency on roaming partners
- Regulatory and country-specific constraints
- Limited visibility into end device behavior

Example: Optional SIM Management Portal

The screenshot displays a web interface for managing a SIM card. At the top, a red SIM card icon is shown next to the device ID '901281' and the label 'Device'. Below this, a tab labeled 'Abonnement' shows the value '901'. A navigation bar contains four tabs: 'Überblick' (selected), 'Bezeichnung', 'Vorgänge', and 'Verlauf'. The main content area is divided into three sections: 'Device-Informationen', 'Letzter Aktivitätsstatus', and 'Standort'. The 'Device-Informationen' section lists: Status (Active.Test), Hersteller (Keine), Device-Modell (Keine), IMEI (zuletzt erkannt) 359 (Zuletzt erkannt am: 21-04-2026 12:18), IMEI (zugewiesen) (Keine), and IMSI. The 'Letzter Aktivitätsstatus' section shows 'Data Session' (Fertig gestellt vor 1 Monat(e), 11 Tag(e); Herunterladen: 0 B. Hochladen: 3.05 KB) and 'Mobilität' (Status: Registriert vor 1 Monat(e), 11 Tag(e); Standortaktualisierung vor 2 Stunde(n), 44 Minute(n), 0 Sekunde(n). Device wird als im Leerlauf angesehen (ist aber möglicherweise nicht erreichbar)). The 'Standort' section features a map of Germany with a location pin. The interface is updated as of 02-06-2026 09:28:16.

901281
Device

Abonnement 901

Überblick Bezeichnung Vorgänge Verlauf

Aktualisiert am 02-06-2026 09:28:16

Device-Informationen

Status
Active.Test

Hersteller
Keine

Device-Modell
Keine

IMEI (zuletzt erkannt)
359
Zuletzt erkannt am: 21-04-2026 12:18

IMEI (zugewiesen)
Keine

IMSI

Letzter Aktivitätsstatus

Data Session
Fertig gestellt vor 1 Monat(e), 11 Tag(e)
Herunterladen: 0 B. Hochladen: 3.05 KB

Mobilität
Status: Registriert vor 1 Monat(e), 11 Tag(e)
Standortaktualisierung vor 2 Stunde(n), 44 Minute(n), 0 Sekunde(n). Device wird als im Leerlauf angesehen (ist aber möglicherweise nicht erreichbar)

Standort

As part of the available support options, a SIM management portal enables customers to conveniently monitor their deployed products.

Connectivity Platform Provider

Objectives:

- Secure communication between device and backend
- Connectivity management (e.g., SIMs, devices)
- Enable scalable IoT deployments

Core Services:

- VPNs and **private APNs**
- Device management platforms
- Data routing and tunneling
- Monitoring and diagnostics

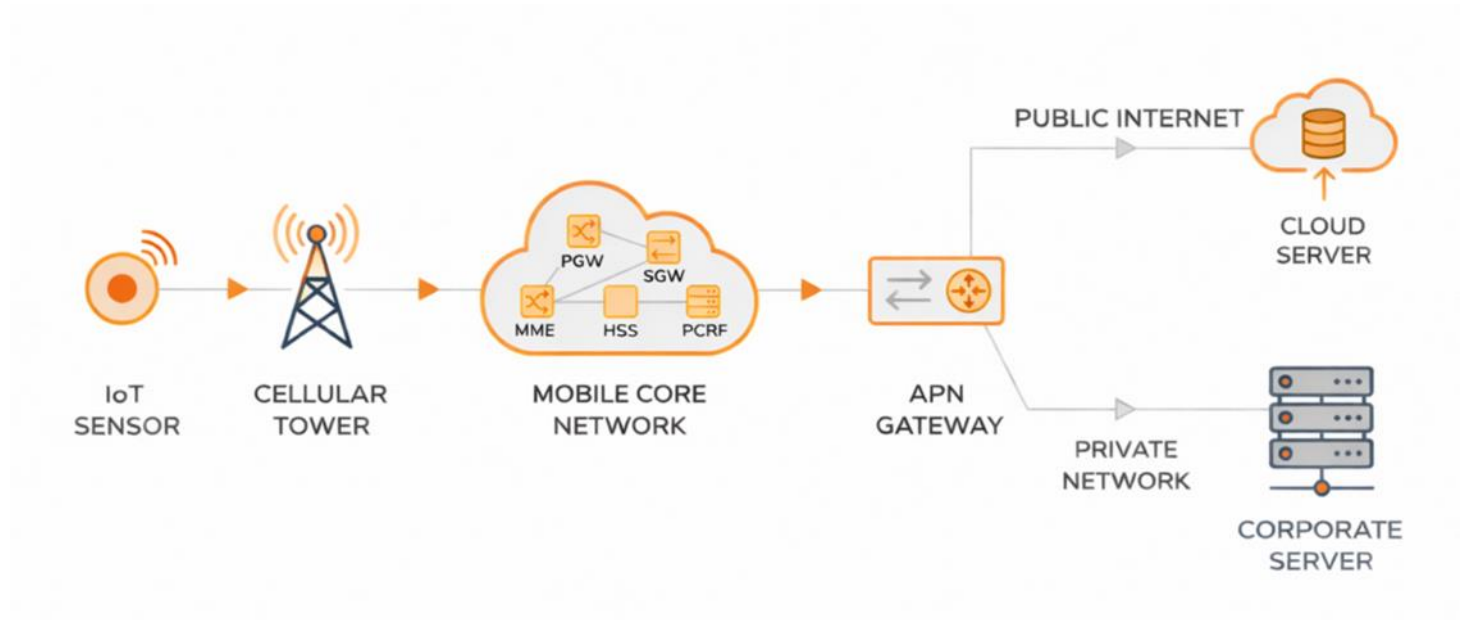
APN and Connectivity Services

What is an APN?

- Entry point from the cellular network into a data network

Key Design Questions:

- Public internet or private APN?
- Static IP required?
- VPN needed?

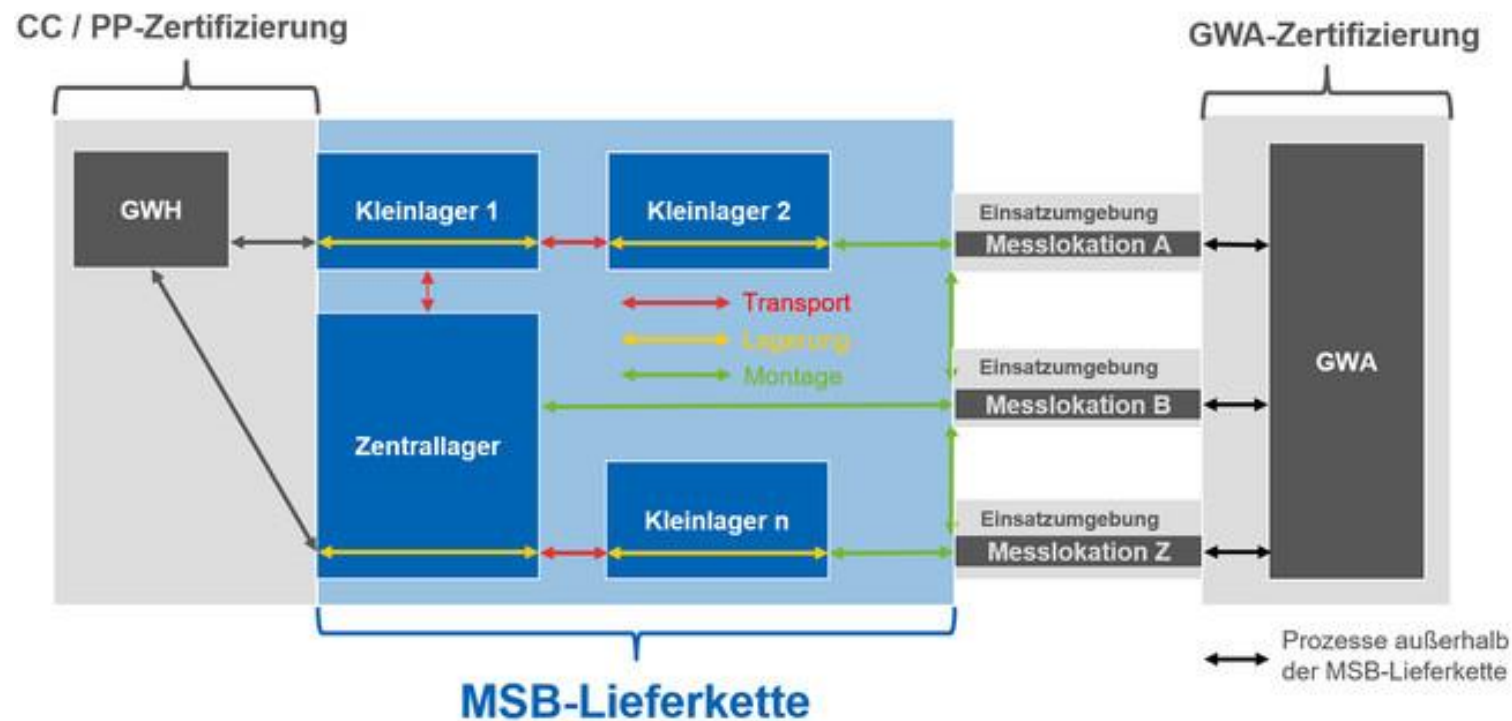


Regulation / Policy Makers

- Requirements (General IoT):
 - Data protection (e.g., GDPR)
 - User consent
 - Data minimization
 - Purpose limitation
- Additional Requirements (Critical Infrastructure)
 - Security certifications (e.g., BSI in Germany)
 - Secure supply chain
 - Compliance and audits

Example: Secure Supply Chain

The BSI mandates a secure supply chain for Smart Meter Gateways (SMGWs) to ensure integrity, authenticity, and protection against tampering throughout production and distribution.



SIM Considerations in IoT

Hardware & Environment

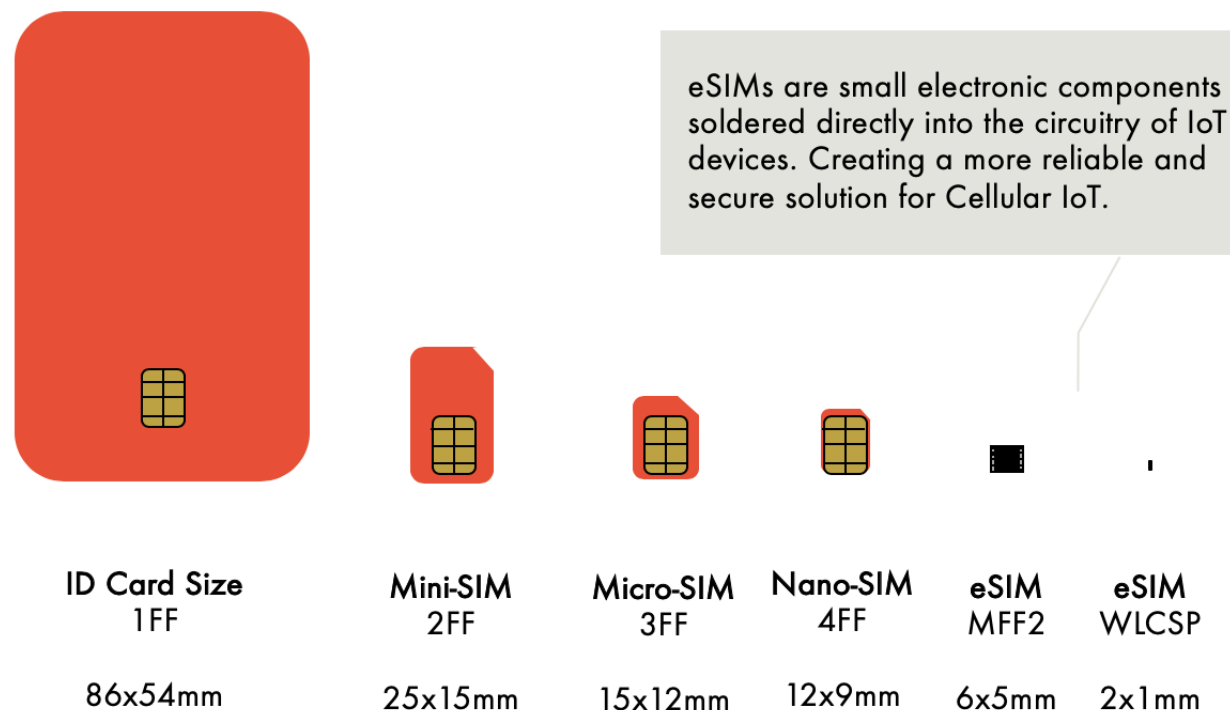
- Physical robustness
- Temperature range
- Device lifetime alignment

Lifecycle Management

- Activation / deactivation
- Remote management
- Blocking in case of misuse

Connectivity & Flexibility

- Operator dependency
- SIM profile switching (eUICC)
- Multi-operator capabilities



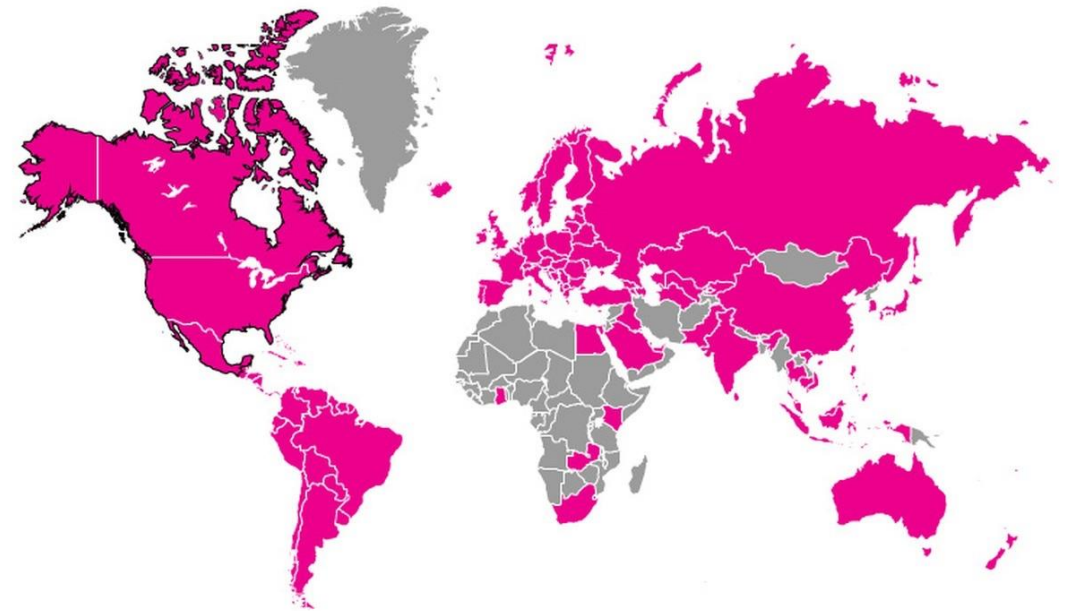
Domestic, International and Roaming

Key Questions for Product Design

- Will the device operate in a single country or globally?
- Is permanent roaming allowed?

Regulatory & Network Considerations

- Are there country-specific regulations?
- Which networks are available in the target region?



 **Mobile Without Borders:** Use your talk, text, & data like you're at home

 **Simple Global:** Unlimited 2G data and texting, plus low, flat-rate calls

*Shows included countries, does not depict coverage.

How to Design a Connectivity Tariff Model

Data Usage

- Uplink vs. downlink volume
- Data per message / transmission
- Frequency (per hour / day / month)

Criticality

- Real-time required?
- Latency constraints
- Reliability requirements

Communication Pattern

- Scheduled? Periodic?
- Event-driven
- On-demand access

Scaling

- Number of devices
- Growth over time
- Geographic distribution

Pricing Example

Category	Item	Price / Condition
Hardware (one-time)	Cellular module	~12 € per device
	SIM card	0 – 2 € depending on form factor
Connectivity	Data plan (option 1)	11 € one-time → includes 8 MB/month per SIM
	Data plan (option 2)	26€/Year → Unlimited, speed-based (70 kbit/s)
Additional Services	VPN per active SIM	~0.08 € / month
	Connectivity platform	~100 € / month
Contract Terms	Duration	3 years
	Volume	10,000 devices per year
	Total volume	30,000 devices
	Minimum commitment	80% of total volume
Conditions	Cancellation fee	8%
	Coverage	EU-wide roaming (all networks)